

Article

Quality Assessment of Generative AI in Cybersecurity Certification

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Abstract

Generative Artificial Intelligence (GenAI), particularly Large Language Models (LLMs), is rapidly changing how higher education approaches teaching, learning, and assessment. In cybersecurity education, professional certification exams are key for measuring competence and helping professionals find better job offers, but there is little research on how GenAI systems perform in these exam settings. This study looks at how three popular LLMs, ChatGPT-5, Gemini-2.5 Pro, and Copilot-2.5 Pro, handle 183 practice questions from the CompTIA Security+ certification. The study used a two-phase evaluation: a domain-based assessment and a full-length practice exam that mirrors real certification tests. The researchers measured model performance with accuracy scores, chi-square tests for statistical differences, and an error taxonomy to spot patterns of mistakes important for education. All three GenAI systems scored above the passing mark, and there were no significant differences between them. Still, the error analysis showed ongoing conceptual and classification mistakes that did not show up in the overall accuracy scores. Our results show that GenAI systems can pass structured certification tests, but accuracy by itself does not fully measure professional skills. The study points out important issues for the reliability and validity of AI-based assessments in higher education and stresses the need for more realistic, concept-focused ways to evaluate GenAI in cybersecurity education.

Keywords: artificial intelligence; assessment; certification; higher education; training



Academic Editors: Amir Ghapanchi,
Reza Ghanbarzadeh and
Purarjomandlangrudi Afrooz

Received: 8 January 2026

Revised: 25 March 2026

Accepted: 25 March 2026

Published: 30 March 2026

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1. Introduction

In the last three years, Generative Artificial Intelligence (GenAI), especially Large Language Models (LLMs), has played a key role in transforming higher education [1–3]. The rapid adoption of these technologies is due to easy access to tools such as ChatGPT, Gemini, and Copilot, which are now used for teaching, learning, administrative tasks,

and decision-making. As a result, LLMs are changing how we think about assessment, teaching methods, and governance, bringing both new opportunities and risks [4,5]. These changes prompt important questions about how institutions can update their teaching and assessment methods.

Research shows that LLMs support many educational tasks [6,7], such as creating content [8,9], tutoring [1], providing feedback [3,10], and helping students prepare for exams [11,12]. More students [2,7,13] are using these tools to learn concepts, solve problems, and prepare for tests, while teachers are finding ways to incorporate them into teaching, assessment, and administration [3,14,15]. However, there are growing concerns about the fairness of assessments, reliance on AI, and the changing role of human expertise, especially when assessments affect professional certification or job opportunities.

Recent studies show that LLMs can match or even outperform humans on standardized and professional exams in fields like medicine [16–19], law [20,21], physics [22–25], and computer science [12,26]. These results challenge the idea that traditional exams always measure skills and knowledge accurately when LLMs are involved. Although cybersecurity is very important, research on how LLMs perform in cybersecurity education and certification remains limited [27–29].

Cybersecurity education stands out in higher education for its reliance on precise technical language, formal standards, and logical reasoning, while also addressing real-world risks and regulations [30–32]. Certifications like CompTIA Security+ help show skills, boost job prospects, build trust in organizations, and prove someone is ready for work. If LLMs can pass these exams, it raises questions for schools and the profession about the value of credentials, the reliability of assessments, and how the field should be regulated.

Most studies on LLMs focus on overall accuracy or benchmark comparisons [23,26,33–37]. These methods provide useful data, but they do not show how or why the models make mistakes, or what these mistakes mean for education [38]. This matters because answers that appear correct but are actually wrong can mislead students more than clearly wrong answers [38]. Knowing the types and patterns of errors is important for assessing the reliability and validity of LLMs.

Recent research in educational assessment shows the need for clear frameworks that connect learning goals, assessment tasks, and the evidence used to judge skills. One common method is Evidence-Centered Design (ECD), which treats assessment as a structured link between skills, evidence, and task design [39]. In this approach, questions are designed to elicit the knowledge or reasoning behind answers. Certification-style questions, like those in professional exams, are designed to help evaluators judge skills based on the accuracy and consistency of the answers.

With this in mind, this study examines how three LLMs, ChatGPT-5, Gemini-2.5 Pro, and Copilot-2.5 Pro, answer 183 practice questions from the CompTIA Security+ certification [40]. The evaluation has two parts: first, it looks at how the models perform across key cybersecurity topics; second, it simulates a full practice exam. The analysis goes beyond accuracy to examine topic coverage, consistency, and patterns in the models' mistakes.

To guide the analysis, this study addresses the following research questions:

RQ1: How accurately do large language models perform across different cybersecurity domains within a certification-based assessment framework?

RQ2: What types of errors do LLMs produce when answering certification-style questions, and how consistent are these error patterns across models?

RQ3: What do the observed performance and error patterns imply for the reliability and validity of AI-assisted assessment in educational and professional certification contexts?

In response to these questions, this study makes three main contributions. First, we provide a domain-based evaluation of model performance using a well-known cybersecurity certification framework, enabling a closer look at strengths and weaknesses in each domain. Second, we create a system for classifying common errors, focusing on how models fail rather than just whether they fail, and providing insight into why these mistakes occur. Third, we discuss what these findings mean for the reliability and fairness of LLMs in creating and evaluating assessments, and consider whether traditional multiple-choice tests remain useful when generative AI tools are readily accessible.

2. Related Works

2.1. Evaluation of LLMs in Exams

Researchers have examined LLM performance on exams and certification tests, particularly in medicine. These studies demonstrate LLMs' capabilities while also highlighting significant shortcomings in reasoning, generalizability, and evaluation design.

In [41], the authors evaluated the performance of nine LLMs on the Chilean Anesthesiology Certification Exam, a key assessment administered in Spanish. The study is notable for its use of Bloom's taxonomy, Item Response Theory, and a thorough error classification system that distinguishes between reasoning, knowledge, and comprehension errors. This methodology delivers a complete understanding of model effectiveness beyond overall accuracy. Results show that LLMs excel at factual recall but show greater error rates on more complex questions that require application and understanding. The study's limitations include a small dataset of 63 questions and a focus on a single exam, both of which limit generalizability. Nevertheless, the research points to the value of integrating cognitive and error-based analyses to identify areas where LLMs struggle with complex reasoning.

In [42], the authors presented MedExpQA, a multilingual benchmark for assessing LLMs on medical questions. A key strength of this benchmark is the inclusion of expert-written explanations, enabling evaluation of both validity and the quality of reasoning. The study identifies persistent challenges, including hallucinations, reliance on outdated information, and significant performance declines in non-English languages. Although MedExpQA advances LLM evaluation, overall performance remains moderate, and integrating external medical knowledge remains difficult. This research highlights the need for clear explanations, multilingual assessment, and robust knowledge integration in LLM evaluation.

In [43], the authors introduced MammoVQA, an extensive benchmark for answering mammography-related questions using both images and text. The primary finding is that leading vision-language systems often perform at chance level on complex diagnostic tasks, raising concerns about their use in clinical practice. A domain-specific model demonstrates improved performance, though the results remain confined to this particular clinical setting. This study shows that strong performance on text-based assessments does not necessarily translate into success on multimodal tasks, underscoring the need for subject-specific evaluation methodologies.

In [44], the authors introduced Priv-IQ, a benchmark designed to assess LLMs on privacy-related competencies, including legal reasoning, multilingual understanding, and visual privacy. The study proposes a novel conceptualization of privacy intelligence and employs an LLM-based evaluator that aligns with human scoring. Although the models perform reasonably well, they continue to struggle with multilingual and domain-specific reasoning. This research is significant because it broadens LLM evaluation past conventional academic assessments and demonstrates the requirement for specialized, diverse testing approaches.

In [45], the authors examined the performance of LLMs and multimodal models on surgical examination questions that involve both text and images, comparing their results with those of human students. A considerable finding is that GPT-4 outperforms students on both question types. However, performance varies considerably across models, particularly for image-based questions, where only one model reaches a passing score. The study is limited to specific exam sections. Yet, it is significant for demonstrating that LLMs can equal or outperform human performance on structured assessments, while also revealing persistent shortcomings in image processing and model consistency.

In [46], the authors assessed several LLMs on the American Board of Anesthesiology BASIC exam using extensive, well-matched datasets. All models achieve scores above the passing threshold, with minimal differences among them. However, the study primarily focuses on accuracy and does not analyze error types or reasoning processes. This finding is important because it demonstrates that LLMs can meet recognized benchmarks, but also suggests that multiple-choice assessments may be insufficient to distinguish between AI and human performance.

In [47], the authors compared the performance of GPT-3.5, GPT-4, and Bard on the NEET-2023 exam. The study examines performance differences among model versions and employs a consensus-based approach to evaluate them. GPT-4 markedly surpasses earlier models, though results continue to vary by subject area. A critical finding is that model agreement does not consistently indicate answer correctness, consequently reducing the reliability of consensus-based evaluation. This research is important as it demonstrates that model agreement is not a strong indicator of accuracy and that performance is subject-dependent.

Taken together, these studies show that LLMs can perform well on exams, but they still have major limitations in reasoning, handling multiple languages, working with both images and text, and evaluation. Many studies focus on overall accuracy and overlook important details, such as the types of errors, the difficulty of the questions, and the validity of the tests. These gaps show that we need more detailed methods for evaluating LLMs, including tests that examine specific fields and error types, as these studies suggest.

2.2. LLMs in Assessment

Researchers have examined how LLMs can support error analysis across different assessment settings. These studies show that LLMs can automate some evaluation tasks, but they also raise reliability concerns across various fields.

In [48], the authors explored how LLMs can assess bone age using radiographic images. The main finding is that LLMs can handle complex, clinically important tasks without specialized training, and their results often match those of experts. The study uses robust statistical methods, such as error metrics (MAE, RMSE) and agreement analysis, to support its approach. However, the models exhibit large error margins (2–3 years) and wide limits of agreement, rendering them unsuitable for clinical use. This is important because it shows that, even when LLMs perform well on average, their error variability and bias remain major problems in high-stakes situations.

In [49], the authors studied how LLMs can assess writing accuracy for Chinese language learners. The main point is a comparison between LLM evaluations and human ratings, along with a detailed error analysis that sorts out where they differ. The study found that models like GPT-4 can be highly precise and perform as well as human evaluators, suggesting they could be useful for large-scale language assessment. Still, the research found differences between human and model judgments, especially with subtle or context-specific errors. This result showed that LLM evaluation standards need to match

those used by humans. The study is important because it shows that error analysis can uncover gaps in model judgment, even when the numbers look good.

In [50], the authors explored how LLMs perform in dietary assessment by analyzing food images and focusing on errors in estimating nutritional values. The main contribution is to show how multimodal LLMs can be used for automated, quantitative assessment, using detailed error metrics such as mean absolute percentage error and bias analysis. The results show that while some models perform as well as traditional self-reported methods, all tend to underestimate and make larger errors for larger portions. This means LLMs can give consistent but biased results, which is a problem for tasks that need high accuracy. The study matters because it shows the value of examining error patterns, such as bias trends, rather than just overall accuracy.

In [51], the authors evaluated LLMs in medical assessment, focusing on patient education for drug-induced liver injury (DILI). The main point is that it combines expert evaluation with a detailed breakdown of errors, including missing information and hallucinations. The study shows that advanced LLMs can be highly accurate and sometimes even outperform some human experts on certain measures. The error analysis finds that missing important information is the most common problem (72%), while hallucinations occur less often but remain important. One limitation is that the study uses a small, specialized set of questions, which may limit the generalizability of the results. This study is important because it shows that understanding the types of errors, not just how often they happen, is key to judging LLM reliability, especially in critical areas like healthcare.

These studies demonstrated that LLMs can support a wide range of assessment tasks, from clinical evaluation to language learning and nutritional analysis, while also revealing consistent challenges, including error variability, systematic bias, and incomplete responses.

3. Materials and Methods

The methodological design of this study considered principles from competency-based assessment and evidence-centered design. The use of certification-style multiple-choice questions reflects established practices for evaluating structured knowledge and domain-specific reasoning, while the combination of domain-level sampling and full-exam simulation supports both fine-grained and holistic performance analysis.

3.1. Study Design

This study uses a system-level evaluation framework to look at how LLMs interact with cybersecurity assessment practices in higher education. The study examines whether LLM performance meets the knowledge, reasoning, and validity standards required for professional cybersecurity certification assessments.

Cybersecurity education involves students, teachers, curricula, certification organizations, and regulations. Here, LLMs serve as support tools that can affect student learning, teacher assessment, and institutional judgments of skills. The study measures model accuracy to check reliability, as well as domain performance, error types, and consistency.

The study's methods reflect real situations in higher education, where students often use AI tools independently to prepare for exams and understand concepts, usually without help from instructors.

3.2. Dataset Source

The dataset used in this study was derived from an official CompTIA Security+ certification preparation book, a licensed resource commonly used in structured training programs for the certification. Unlike publicly accessible question banks or informal online repositories, this source provides curated and domain-aligned content that reflects the

scope, structure, and level of difficulty expected in certification assessments. To ensure the ethical use of the material and avoid potential interference with active certification content, the study used questions from a previous version of the Security+ certification framework. This approach preserves the integrity of current certification materials while maintaining alignment with established cybersecurity knowledge domains.

The use of certification-oriented content enables evaluation within a controlled and standardized assessment context, where questions are explicitly designed to measure structured reasoning, conceptual understanding, and familiarity with established security principles. While it is not possible to completely exclude the possibility of partial overlap with LLM training data, the use of licensed, non-public materials reduces the likelihood of direct exposure compared to openly available sources. Moreover, the study emphasizes error patterns, response consistency, and domain-level performance, providing insights that extend beyond simple answer recall. This sampling strategy aligns with established assessment design approaches that distinguish between domain-specific measurement and overall exam-level performance, supporting both validity and interpretability of results

3.3. Question Selection

The question dataset was organized according to the six primary domains defined in the CompTIA Security+ certification framework:

1. Threats, Attacks, and Vulnerabilities.
2. Technologies and Tools.
3. Architecture and Design.
4. Identity and Access Management.
5. Risk Management.
6. Cryptography and Public Key Infrastructure.

To ensure balanced domain representation while maintaining manageable experimental conditions, a structured sampling strategy was applied. From each domain-specific question pool, a subset of items was selected to cover the principal cybersecurity topics outlined in the certification objectives. In most domains, 15 questions were selected, while Domain 4 contained fewer available items and therefore contributed 8 questions. This resulted in a total of 83 domain-specific questions.

To complement the domain-based evaluation, an additional 100 questions were selected from a full-length practice examination within the same source. This component was designed to simulate the structure and difficulty of a complete certification-style test, allowing the models to be evaluated under more realistic assessment conditions.

In total, the dataset consisted of 183 multiple-choice questions. This combination of domain-focused sampling and full-exam simulation enables the evaluation of both topic-level understanding and overall performance in a certification-like testing environment. The correctness of the selected questions and answers was verified against the preparation materials and the official certification domain objectives to ensure alignment with established cybersecurity concepts.

3.4. Evaluation Phases

The evaluation was conducted using a two-phase design to capture domain-level behavior and overall assessment performance.

3.4.1. Phase 1: Domain-Based Evaluation

Phase 1 analyzed 83 multiple-choice questions distributed across the six Security+ domains. This phase focused on domain sensitivity, conceptual comprehension, and consistent reasoning. The structure facilitated direct comparison of how LLMs respond

to various types of cybersecurity knowledge, spanning from abstract risk assessment to protocol-specific details.

3.4.2. Phase 2: Full Practice Examination

Phase 2 employed a comprehensive 100-question practice examination meticulously designed to replicate the conditions of the certification exam. The examination comprised single-answer questions, multiple-answer items, and scenario-based problems. This phase evaluated overall proficiency, response consistency, and endurance within an extended assessment environment.

Together, the two phases facilitated the analysis of LLMs' performance within modular learning configurations and environments characterized by exam-length evaluations.

3.5. Interaction Conditions

The study tested the following LLMs: ChatGPT-5, Gemini-2.5 Pro, and Copilot-2.5 Pro. ChatGPT-5 is developed by OpenAI, based in San Francisco, United States; Gemini-2.5 Pro is produced by Google DeepMind, headquartered in Mountain View, United States; and Copilot-2.5 Pro is created by Microsoft, located in Redmond, United States. These tools are commonly available to students and teachers in higher education.

Each model received the same inputs under standard conditions:

- The same questions;
- In the same order;
- With identical formatting;
- Without additional contextual prompts or follow-up queries.

Only the first response for each question was recorded. The evaluation did not allow new attempts, clarifications, or repeated tries. Each question was submitted as a standalone prompt, containing the full multiple-choice question and its answer options, exactly as written in the dataset. No additional context, instructions, prompt engineering strategies, or follow-up queries were used. The question text itself, therefore, served as the prompt, reflecting typical student interactions with publicly available LLM interfaces during exam preparation. The complete question dataset and response logs used in the evaluation are provided as Supplementary Materials to support transparency and procedural replication. The models were accessed through their public web interfaces via free options, making the study relevant to real educational practice.

All evaluations were conducted between 14 October 2025 and 19 October 2025, during which the same model interfaces and deployment versions were available. Each interaction followed an identical prompt structure in which the question text was provided without additional context, instructions, or prompting strategies. The full set of prompts and response logs used in the evaluation is documented in the study materials to support transparency and facilitate procedural replication.

3.6. Source of Questions

The practice questions used in this study were derived from officially published CompTIA Security+ certification preparation materials, which are not freely or publicly accessible and require authorized purchase. These materials form part of established certification training resources and are designed to align closely with the official exam objectives. To further mitigate potential data leakage and preserve the integrity of current certification content, this study deliberately employed materials from a previous version of the certification. This approach ensures that the evaluation neither interferes with nor exposes current exam questions while remaining aligned with standardized cybersecurity knowledge domains.

3.7. Accuracy Measurement

The study measured accuracy as the proportion of correctly answered questions relative to the total number of questions presented to each model. Each response received a binary score, correct or incorrect, based on alignment with the officially correct answer.

Accuracy was calculated at two complementary levels:

- Overall accuracy, which shows performance on all questions;
- Domain-level accuracy, which shows performance within each Security+ domain.

In addition to accuracy metrics, the study conducted a qualitative analysis of errors. Incorrect responses were reviewed to identify systematic patterns, including:

- Confusion between closely related cybersecurity concepts;
- Incorrect application of cryptographic workflows;
- Misinterpretation of access control models;
- Protocol and port-level inaccuracies;
- Reliance on modern standards while overlooking legacy specifications.

3.8. Statistical Analysis

The analysis used descriptive and inferential statistical methods to examine consistency in performance across models. Chi-square tests were initially used to examine overall differences in accuracy distributions between models. Because all models responded to the same set of questions, additional paired item-level comparisons were conducted using McNemar's test to account for the dependence among observations.

Effect sizes were calculated using Cramer's V to evaluate the practical relevance of observed differences. Statistical results supported a system-level interpretation over model superiority, emphasizing assessment stability and reliability over ranking.

3.9. Error Pattern Analysis and Formal Taxonomy

To move beyond surface-level accuracy, the study conducted a structured error analysis using a formal, assessment-oriented taxonomy. Each incorrect response generated by a model was treated as a separate error instance, reflecting real assessment conditions where each response independently affects evaluation outcomes. Based on observed mistakes, errors were classified into three categories:

- Conceptual and classification errors;
- Protocol and technology selection errors;
- Low-level technical recall errors.

This taxonomy emphasizes assessment validity rather than model implementation details. Error patterns were interpreted as indicators of systemic risk in AI-enhanced assessment environments, particularly when plausible but incorrect answers may reinforce misconceptions.

3.10. Human Oversight

To support responsible AI use in higher education, the evaluation included human oversight. Cybersecurity instructors and experts reviewed error logs to identify error patterns, explain technical errors, and assess their educational significance.

This approach recognizes that LLM outputs need expert review to guide assessment design, curriculum development, and teaching decisions. It also supports the idea that LLMs can help with learning and assessment, but do not replace human judgment when evaluating professional skills.

3.11. Workflow of the Methodology

Figure 1 shows the overall workflow for evaluating LLM performance in a cybersecurity certification assessment. This workflow follows the study's framework, which integrates standardized testing, accuracy metrics, and error analysis.

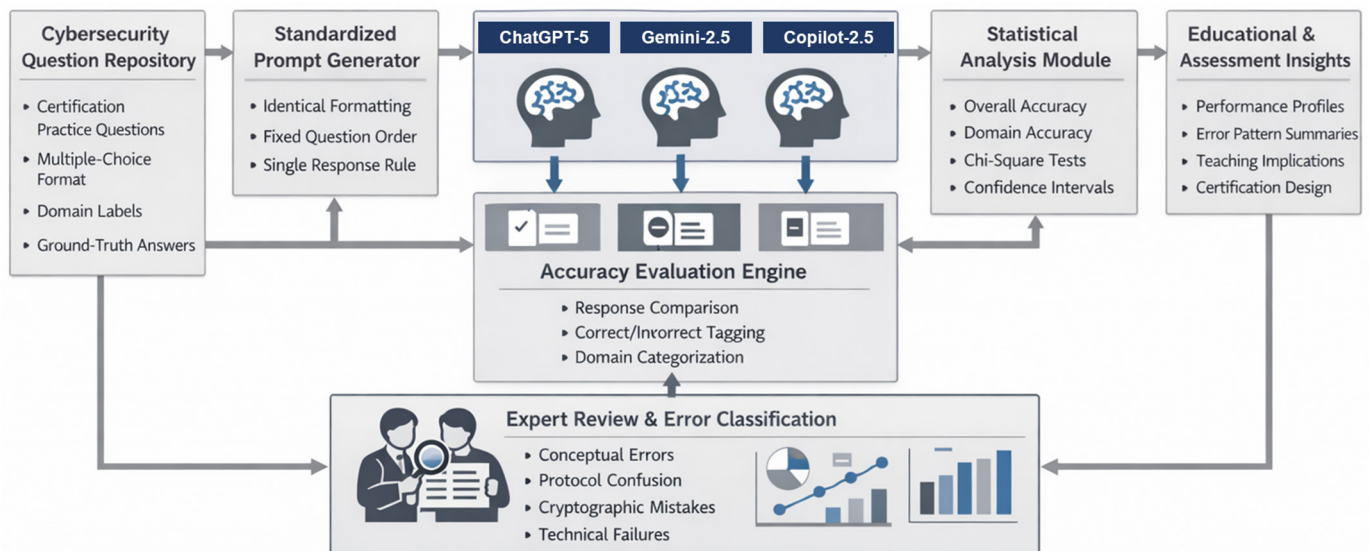


Figure 1. Methodological workflow used in this study.

The process starts with a set of multiple-choice practice questions for cybersecurity certification, all based on CompTIA Security+ domains. Each question has a confirmed correct answer and a domain label to match professional certification standards.

A prompt generator formats each question in a set way, uses controlled wording, and allows only one response. This ensures all models receive the same input and reflects real exam preparation, where there is no chance to ask follow-up questions.

The formatted questions are submitted independently. The formatted questions are sent separately to ChatGPT-5, Gemini-2.5 Pro, and Copilot-2.5 Pro. Each model provides one answer per question, and these answers are recorded as-is, without any changes or follow-up. A search engine, which compares responses against ground-truth answers, assigns binary correctness labels, and aggregates results at both overall and domain levels. These outputs feed into a statistical analysis module that computes descriptive accuracy metrics and applies chi-square tests to examine performance differences across models, supporting system-level interpretation rather than competitive ranking.

At the same time, experts review all incorrect answers and classify the errors using a cybersecurity error system. This step helps identify common mistakes in concepts, protocol choices, and technical details, revealing limitations that accuracy scores might miss. In the final step, the study combines the numbers and expert findings to provide insights for education and assessment. These results help guide discussions about the reliability and validity of the assessments, how AI can support learning, and how to design better cybersecurity tests for AI-supported classrooms.

4. Results

This section shares the results from evaluating three LLMs, ChatGPT-5, Gemini-2.5 Pro, and Copilot-2.5 Pro, using cybersecurity certification practice questions. Instead of comparing their performance, the findings aim to show how LLMs operate in AI-supported assessments in higher education, focusing on reliability, validity, and error patterns.

4.1. Evaluation Design

The evaluation consisted of two complementary phases that reflect different levels of cognitive and professional demands in cybersecurity education.

4.1.1. Domain-Based Evaluation

Phase 1 focused on domain-level reasoning and conceptual understanding. The test included 83 multiple-choice questions distributed across the six official Security+ domains. The question distribution followed the official exam structure. Domains 1, 2, 3, 5, and 6 each included fifteen questions, while Domain 4 included eight questions. This structure ensured realistic coverage while preserving balance across knowledge areas.

4.1.2. Full Practice Examination

Phase 2 simulated a complete certification scenario using a 100-question practice exam. The exam included single-answer multiple-choice items, multiple-answer questions, and scenario-based problems that required applied reasoning. This phase evaluated overall proficiency, response stability, and performance consistency over an extended assessment period.

Together, the two phases enabled analysis at both modular and system levels, reflecting typical learning and certification contexts.

4.2. Accuracy as an Indicator of Assessment Reliability

Accuracy was measured as the percentage of correct answers among all questions. This was calculated for each domain and for each system, as shown in Equation (1).

$$\text{Accuracy} = \frac{\sum_{i=1}^n T_i}{n} \times 100\% \quad (1)$$

where

n is the total number of questions;

$T_i = 1$ if the model's response matched the official answer, and $T_i = 0$ otherwise.

In this study, accuracy is interpreted as an operational indicator of assessment reliability within a multiple-choice testing context, rather than as a proxy for comprehensive domain expertise. While accuracy provides a useful baseline for comparing model performance, it does not capture underlying reasoning processes or the nature of incorrect responses. Therefore, all incorrect answers were retained for subsequent error pattern analysis, enabling a more comprehensive evaluation of potential threats to assessment validity.

4.3. Phase 1 Results: Domain-Based Assessment Performance

Table 1 presents the results of the domain-based evaluation (83 questions). All three models achieved high levels of accuracy, with Gemini-2.5 Pro scoring 89%, ChatGPT-5 87%, and Copilot-2.5 Pro 86%.

Table 1. Domain-based results (83 questions).

Domain	ChatGPT-5	Copilot-2.5 Pro	Gemini-2.5 Pro
Threats, Attacks, and Vulnerabilities	13	12	13
Technologies and Tools	12	12	12
Architecture and Design	13	14	14
Identity and Access Management	6	6	6
Risk Management	13	12	14
Cryptography and PKI	15	15	15
Total Correct	72	71	74
Accuracy	87%	86%	89%

The models demonstrated particularly strong performance in Architecture and Design, Risk Management, and Cryptography and Public Key Infrastructure, domains that emphasize conceptual reasoning and structured decision-making. This suggests that LLMs are well aligned with higher-level conceptual learning objectives commonly found in cybersecurity education.

Performance in Technologies and Tools and Identity and Access Management was also consistent across models, indicating competence in operational and applied knowledge areas. Additionally, all models successfully interpreted a command-line-based question, suggesting emerging capabilities in handling multimodal or context-rich inputs, which are increasingly relevant in cybersecurity assessment.

4.4. Phase 2 Results: Full Practice Examination Performance

Table 2 summarizes the results of the full 100-question practice examination. All models exceeded typical Security+ passing thresholds, with Gemini-2.5 Pro achieving 97%, Copilot-2.5 Pro 96%, and ChatGPT-5 93%.

Table 2. Full practice exam results (100 questions).

Model	Correct	Accuracy
ChatGPT-5	93	93%
Copilot-2.5 Pro	96	96%
Gemini-2.5 Pro	97	97%

The findings indicate that LLMs can perform consistently well under summative, certification-like conditions, in which questions are presented in a continuous, mixed-domain format. This raises important considerations for assessment design, particularly regarding the ability of traditional multiple-choice formats to differentiate between human learners and AI-assisted responses in environments where such tools are accessible.

4.5. Combined Performance Across Assessment Phases

Table 3 presents the aggregated results across all 183 questions. All models achieved overall accuracy above 90%, with Gemini-2.5 Pro at 93.44%, Copilot-2.5 Pro 91.26%, and ChatGPT-5 90.16%.

Table 3. Combined results (183 questions).

Model	Correct	Incorrect	Overall Accuracy
ChatGPT-5	165	18	90.16%
Copilot-2.5 Pro	167	16	91.26%
Gemini-2.5 Pro	171	12	93.44%

While these results indicate strong overall performance, accuracy alone provides an incomplete evaluation. High aggregate scores may obscure systematic patterns of error, particularly when incorrect responses are plausible or contextually coherent. Such patterns have direct implications for the validity and interpretability of AI-assisted assessment systems, reinforcing the need for complementary analyses beyond aggregate performance metrics.

4.6. Statistical Analysis of Model Performance

Statistical analysis was conducted to determine whether observed differences in model performance were statistically significant and to assess whether variations could be attributed to systematic effects rather than random variation. Chi-square tests of indepen-

dence were applied to categorical outcome data (correct vs. incorrect responses) to compare performance distributions across models. Results indicated no statistically significant differences in accuracy between models in any evaluation phase:

- Phase 1 (83 questions): $\chi^2(2) = 0.50$, $p = 0.78$.
- Phase 2 (100 questions): $\chi^2(2) = 1.33$, $p = 0.51$.
- Combined dataset (183 questions): $\chi^2(2) = 1.95$, $p = 0.38$.

Effect sizes, measured using Cramér's V , ranged from 0.045 to 0.081, indicating negligible to very small effects. These results suggest that differences in aggregate accuracy between models are minimal and not statistically meaningful.

To account for the paired nature of responses across identical questions, McNemar's tests were additionally conducted at the item level. These analyses produced consistent results, confirming that differences in model responses were not statistically significant when evaluated on a per-question basis.

It is important to note that the Chi-square test was applied to binary outcome distributions rather than aggregated accuracy scores, ensuring methodological appropriateness. However, this approach has limitations, including the potential loss of item-level variability and dependence between observations. For this reason, statistical testing is complemented by domain-level analysis and structured error classification, which provide deeper insight into model behavior.

From an assessment perspective, these findings indicate that model selection does not substantially influence outcomes in certification-style multiple-choice tests. Instead, the results highlight the importance of assessment design, particularly in contexts where generative AI systems are readily available to learners.

4.7. Error Pattern Analysis

To complement accuracy-based evaluation and address potential threats to assessment validity, this study conducted a structured analysis of error patterns in all incorrect responses. Rather than treating errors as isolated outcomes, the analysis examines systematic patterns in model failures, providing insight into how and why incorrect answers occur in certification-style assessments. Each incorrect response was treated as a distinct analytical unit, consistent with standard assessment conditions in which each question contributes independently to overall performance. Error analysis was conducted across all domains in which incorrect responses were observed (Domains 1–5). Domain 6 (Cryptography and Public Key Infrastructure) was excluded from this analysis because it contained no errors, indicating uniformly high performance in that area. This approach enables the identification of recurring failure modes that are not visible through aggregate accuracy metrics, thereby supporting a more comprehensive evaluation of LLM reliability in educational assessment contexts.

4.7.1. Formal Error Taxonomy

A domain-informed error taxonomy was developed through an iterative review of incorrect responses. The classification process focused on identifying patterns that are pedagogically meaningful and relevant to certification assessment, rather than on model-specific or implementation-level errors. This approach enables the analysis to reflect how errors relate to learning outcomes and assessment constructs, rather than internal model behavior. Table 4 summarizes the error categories identified.

This taxonomy is designed to align error analysis with learning objectives and assessment constructs, enabling interpretation in terms of conceptual understanding, applied knowledge, and factual recall. By abstracting from individual questions to broader categories, the framework supports comparability and reproducibility across similar studies.

Table 4. Formal cybersecurity error taxonomy.

Code	Error Category	Description
E1	Conceptual errors	Incorrect categorization of threats, malware types, access control models, risk processes, or security principles
E2	Protocol and technology selection errors	Confusion between protocols, tools, or technologies with overlapping functions but distinct technical properties
E3	Low-level technical recall errors	Incorrect recall of ports, protocol numbers, or configuration-level details

4.7.2. Error Distribution

Across Domains 1 to 5, a total of 31 errors were identified. Table 5 shows the distribution of errors.

Table 5. Distribution of error categories.

Error Category	Error Count	Accuracy
Conceptual errors	18	58.06%
Protocol and technology selection errors	10	32.26%
Low-level technical recall errors	3	9.68%
Total	31	100%

This distribution indicates that the majority of errors are associated with misinterpretation of conceptual boundaries and relationships, rather than failures of factual recall. While LLMs demonstrate strong performance in retrieving discrete technical information, they are more prone to errors when tasks require distinguishing between closely related concepts or applying knowledge in context.

From an assessment perspective, this finding is particularly significant. Conceptual errors often produce responses that are plausible and contextually coherent, increasing the risk that such errors may go undetected by learners. This has direct implications for the validity of AI-assisted assessment systems, as high overall accuracy may mask underlying weaknesses in conceptual understanding.

Moreover, the relatively low frequency of recall-based errors suggests that traditional multiple-choice assessments, which often emphasize factual knowledge, may overestimate model competence when used as the sole evaluation method. These results reinforce the need for assessment designs that better capture higher-order reasoning and conceptual differentiation, particularly in environments where LLMs are readily accessible.

5. Discussion

Evaluating ChatGPT-5, Gemini-2.5 Pro, and Copilot-2.5 Pro with 183 cybersecurity certification questions reveals their strengths and limitations in assessment. Although all three systems performed well with structured questions, the types of errors observed raise concerns about the reliability of these assessments and how best to integrate such tools into professional training. Instead of a direct comparison, this discussion explores the broader implications for AI-supported assessment. The main issue is not just whether LLMs can answer certification questions, but how their use changes core assumptions about learning, evaluation, and academic integrity in cybersecurity education.

5.1. Demonstrated Strengths

Throughout all phases of testing, the three systems achieved consistently high scores. Accuracy ranged from 86% to 89% across domains, and overall exam scores exceeded 93%. These outcomes surpass the typical passing threshold for CompTIA Security+, indicating that current LLMs are well aligned with existing assessment formats. Performance was strongest in domains requiring structured reasoning, including Architecture and Design, Risk Management, and Cryptography and PKI. These areas require understanding abstract concepts, policy, and defined security principles. The lack of errors in Cryptography and PKI shows that content based on established standards is handled consistently. The systems also handled questions that combined visual and textual elements, such as command-line screenshots. This is important because cybersecurity often involves interpreting logs, dashboards, and system outputs. Instructionally, this allows for more authentic tasks. In this context, LLMs can support practice, provide feedback, and guide exploration, especially for introductory and intermediate learners.

5.2. Limits of Precision

Despite strong overall results, the error patterns reveal underlying weaknesses. Incorrect answers were most common in areas needing subtle conceptual distinctions and precise definitions. Many errors involved confusing related concepts, such as mistaking whaling for spear phishing, misclassifying malware, or mismatching access control models with their definitions. These mistakes are difficult to spot quickly because they often seem plausible. Additional issues arose during protocol and technology selection. The systems often confuse tools with similar functions but different properties, such as TACACS, TACACS+, and RADIUS. This suggests difficulty in accurately applying formal standards, which are essential in cybersecurity practice and certification. Errors involving simple factual recall were rare, suggesting most failures stemmed from conceptual ambiguity rather than missing information. Performance varied by domain; no errors occurred in Cryptography and PKI, but results were weaker elsewhere. This inconsistency is important for assessment.

5.3. Implications of Potential Training Data Exposure

A possible explanation for the high scores is that the models may have encountered similar questions during training, especially given how widely Security+ preparation materials circulate online. Even so, the results suggest that simple memorization is not enough to explain what we observed. The models made a noticeable number of mistakes (31 in total), many of them repeating the same kinds of conceptual misunderstandings. If their answers were mainly drawn from memorized question–answer pairs, we would expect performance to be almost flawless. Instead, most errors occurred where concepts overlap or need to be clearly distinguished, rather than in recalling specific facts. This points to difficulties in applying definitions and standards rather than gaps in stored information. The fact that all three systems showed similar patterns of mistakes also suggests common ways of processing information, rather than reliance on retrieving exact answers.

Examining these errors more closely helps us understand how the models work. Most mistakes occur in areas where it is important to distinguish among similar security principles, protocols, or threat types, rather than in recalling facts. This shows that while the models usually remember technical details well, they are less reliable when they need to use that knowledge accurately. The fact that all three systems have similar problems suggests these are common weaknesses in dealing with specific concepts, not just random memory errors.

5.4. Implications for AI-Enhanced Assessment

These findings prompt a reconsideration of how learning is evaluated when LLMs are easily accessible. If these systems can pass certification-style exams with ease, traditional assessments may no longer clearly demonstrate individual competence. These findings do not make assessment irrelevant, but they do require adaptation. Scenario-based tasks, explanation-focused questions, hands-on labs, and interactive simulations allow for more thorough evaluation of applied knowledge and judgment. These formats depend less on recognition and better reflect real-world practice. The results also raise concerns about academic integrity. When correct answers can be generated quickly, the focus may need to shift from detection to assessment designs that require justification, transparency, and accountability. This approach promotes integrity through thoughtful design rather than restrictive measures.

5.5. Institutional Responsibility

The study highlights the growing importance of AI literacy. In cybersecurity education, students need more than technical skills; they must critically evaluate AI-generated responses, identify recurring errors, and verify information using reliable sources. Teaching roles are also evolving. Instructors are now more responsible for shaping learning environments, interpreting assessment results, and guiding the responsible use of AI. Institutions must establish clear policies that support transparent and equitable use of LLMs and address ethical considerations.

5.6. Broader Educational Implications

LLMs influence both task completion and knowledge assessment. In cybersecurity, where errors can have serious consequences, maintaining trust in certification and evaluation is essential. The results show that high accuracy does not always reflect deep competence. Addressing this gap requires changes to assessment design, more focus on AI literacy, and continued reliance on human judgment. If LLMs are used as part of a broader learning environment, rather than as substitutes for instruction or assessment, their benefits can be realized without compromising the credibility of professional qualifications. Although this study focuses on cybersecurity certification, the findings have broader implications for assessment practices in other disciplines that rely on structured testing formats. Similar challenges related to LLM performance, assessment validity, and interpretation of accuracy metrics are likely to emerge in fields such as computer science, health education, and engineering. The proposed error analysis framework may therefore support cross-disciplinary evaluation of LLM behavior in educational contexts.

6. Conclusions

Generative AI is transforming teaching, learning, and assessment in higher education. This study analyzed the performance of three popular systems, ChatGPT-5, Gemini-2.5 Pro, and Copilot-2.5 Pro, on a professional cybersecurity certification exam. Using 183 CompTIA Security+ practice questions and a clear evaluation method, the analysis shows both the strengths and weaknesses of LLMs on certification-style tests. All three systems scored well, meeting or passing the usual standards, with no major differences between them. In multiple-choice tests, the LLMs were accurate across a range of cybersecurity topics. However, their mistakes reveal clear weaknesses, as many errors stem from misunderstandings or misuse of basic cybersecurity concepts. These results suggest that high test scores do not always indicate original understanding or sound reasoning, and accuracy alone is insufficient to assess reliability. These results suggest we need to rethink how we assess students when AI tools are available. Multiple-choice tests may no longer clearly

distinguish between a student's own knowledge and AI-assisted answers. While this study did not test other formats, the findings indicate that assessments should more effectively measure reasoning, judgment, and the application of knowledge in real-world situations. Studying other datasets, more certification programs, and comparing results with human performance would help us better understand how reliable these systems are and what they mean for education.

Limitations

There are a few important limitations to consider with these findings. First, the dataset is based only on the CompTIA Security+ certification, so the results might not apply to other fields, certification systems, or educational settings. Although this certification provides a clear, widely recognized standard, other areas may have different knowledge requirements, challenges, and testing styles, which could affect how LLMs perform. Second, the dataset contains 183 questions, which is a small number compared to those in large benchmark studies, even though the questions were carefully selected. They cover several areas of the certification, but not every possible topic, difficulty, or question type. So, these results should be viewed as general trends rather than definite proof of the model's performance in every situation. Third, the study uses multiple-choice questions that primarily assess recognition, conceptual understanding, and the selection of correct answers. These questions do not measure higher-level skills like problem-solving, hands-on work, or complex decision-making. As a result, the findings reflect only part of what it means to be skilled in cybersecurity. Fourth, although the dataset came from licensed, non-public certification prep materials, it is still possible that similar content appeared in LLM training data. This represents a broader methodological challenge in evaluating LLMs on domain-specific knowledge tasks and should be considered when interpreting high accuracy results. Finally, the study does not compare the model's results to human performance or random guessing, which would help put the findings in context. Future research should include these comparisons and use larger, more varied datasets to strengthen the evaluation and make it more useful.

Supplementary Materials: The complete question dataset and response log can be downloaded at: <https://www.mdpi.com/article/10.3390/informatics13040053/s1>, Table S1: Selection of questions; Table S2: LLMs answers to each question; Table S3: Correct and incorrect answers in the 100 questions.

Author Contributions: Conceptualization, V.G.F., R.O., L.J.M. and R.M.-P.; methodology, V.G.F., R.O., L.J.M., H.T.-C., P.V.-A. and R.M.-P.; software, A.O.-B., A.G.-P. and R.A.F.-C.; validation, H.T.-C., P.V.-A., J.A.L.-B. and R.M.-P.; formal analysis, H.T.-C., P.V.-A. and R.M.-P.; investigation, V.G.F., R.O. and L.J.M.; resources, H.T.-C. and R.M.-P.; writing—original draft preparation, V.G.F., R.O., A.O.-B., A.G.-P., R.A.F.-C., J.A.L.-B. and R.M.-P.; writing—review and editing, L.J.M., H.T.-C. and P.V.-A.; funding acquisition, R.M.-P. All authors have read and agreed to the published version of the manuscript.

Funding: This research was partially funded by the UNIVERSIDAD CATOLICA DEL NORTE through the Concurso Fondo de Desarrollo de Proyectos Docentes de Pregrado (FDPD 2023), under grant number DGPRES.N° 176/2023.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article/Supplementary Materials. Further inquiries can be directed to the corresponding author.

Acknowledgments: During the preparation of this study, the authors used Grammarly Pro (v1.2.189.1739) for the purpose of improving the quality of writing.

Conflicts of Interest: The authors declare no conflicts of interest.

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